

Terry balance wheel skeleton clock from the 19th century



Using weights

- **What?** Balance wheel
- **Who?** Unknown
- **Where and when?** Europe, 17th century

The balance wheel is a weighted device that swings back and forth, winding and unwinding a spring in a constant cycle. Each swing of the balance wheel creates a tick or a beat and drives the gears that make the hands move one position.

A more accurate clock

- **What?** Pendulum clock
- **Who?** Christiaan Huygens
- **Where and when?** The Netherlands, 1657

A physicist, astronomer, and mathematician, Christiaan Huygens also invented the pendulum clock. Until then, the best clocks were accurate to about 15 minutes a day, whereas the pendulum clock was accurate to about 15 seconds.

Model based on the design of Huygens's first pendulum clock



Electric watch

- **What?** Hamilton Electric 500
- **Who?** Hamilton Watch Company
- **Where and when?** US, 1957



The Hamilton Electric 500 was the first battery-operated electric wristwatch and the first never to need winding. It was marketed as the watch of the future and issued in several suitably modern designs, including the shield-shaped Ventura (left).

Hamilton "Ventura" electric watch



Digital watch

- **What?** Hamilton Pulsar P1
- **Who?** Hamilton Watch Company
- **Where and when?** US, 1972

Luxury watch brand Hamilton was the first to release a digital display watch, in 1972. It was ridiculously expensive, but then the case was 18 carat gold. Cheaper, mass-market versions soon followed.



Seiko 06LC digital watch from 1973

Smartwatch

- **What?** TrueSmart
- **Who?** Omate
- **Where and when?** US, 2013

The first smartwatch to have all the functions of a smartphone was the TrueSmart, which appeared in early 2013. Since then, many larger companies have launched smartwatches, including Samsung, Sony, and Apple.



Early wristwatches were designed as jewelry for women. The first was probably made for an aristocrat some time in the mid-19th century.